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Exercise: Build a reproducible malaria report

Swiss TPH Quarto workshop

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1 Aim

In this exercise, you will turn the minimal `report-starter.qmd` document into a reproducible report. You will add Markdown text, executable R code, a figure, a table, captions, and cross-references.

2 Before you begin

1. Open the `exercises` folder as your working folder.
2. Open `report-starter.qmd` in RStudio.
3. Replace `Your Name` in the YAML header with your name.
4. Render the document once. A simple HTML document should be created without errors.

You can switch between the **Source** and **Visual** editors in RStudio using the button at the top-left of the editor pane.

3 Tasks

3.1 Try another output format

The `format` field in the YAML header controls the output type. Render the starter as HTML first, then try changing the value to `typst` or `docx`.

For a Typst PDF with a table of contents and numbered sections, use:

```yaml
format:
  typst:
    toc: true
    number-sections: true
    mainfont: "Arial"
```

Return to the format you want to use for the rest of the exercise.

3.2 Load the packages and data

Immediately below the YAML header, insert an R code chunk using RStudio's **Insert** button (or **Code > Insert Chunk**), then put the following content inside it. The `include: false` option means that the code runs, but neither the code nor its output appears in the rendered report.

```
#| label: setup
#| include: false

library(readr)
library(dplyr)
library(ggplot2)
library(knitr)

cases <- read_csv("data/cases.csv", show_col_types = FALSE)
```

Render again to check that the packages load and the data file can be found.

3.3 Add headings and formatted text

Add these headings below the setup chunk:

```
# Introduction

# Results

## Cases over time

## Summary table

# Discussion
```

Under **Introduction**, write two or three sentences describing the report. Include at least:

- one **bold** phrase, written as `**bold**`;
- one *italic* phrase, written as `*italic*`; and
- one link, for example `[Quarto](https://quarto.org)`.

Render the document and inspect how the headings and formatting appear.

3.4 Add inline R code

Numbers derived from data should not be typed by hand. Add the following sentence under your introduction and replace both bracketed placeholders with inline R code:

> The dataset covers [number of rows] monthly observations and [sum of `n`] cases in total.

Inline R code is enclosed in backticks and starts with the letter `r` followed by a space and an expression. For the first value, put `r nrow(cases)` between backticks. For the second, put `r sum(cases$n)` between backticks. Render and check that the sentence contains numbers rather than code.

3.5 Create a figure with a caption

Under **Cases over time**, insert another R code chunk and put the following content inside it:

```
#| label: fig-cases
#| fig-cap: "Monthly reported malaria cases by region."

ggplot(cases, aes(month, n, colour = region)) +
  geom_line() +
  labs(x = "Month", y = "Cases")
```

The `fig-` prefix makes the labelled figure cross-referenceable. Change the caption if you can make it more informative, then render the report.

3.6 Experiment with chunk options

Add the following options to the figure chunk one at a time. Render after each change so that you can see its effect.

```
#| echo: false
#| warning: false
```

- `echo: false` hides the source code but retains its output.
- `warning: false` suppresses warnings in the rendered document.

3.7 Create a summary table

Under **Summary table**, insert another R code chunk and put the following content inside it:

```
#| label: tbl-summary
#| tbl-cap: "Total and mean monthly cases by region."

cases |>
  group_by(region) |>
  summarise(total = sum(n), mean = round(mean(n), 1)) |>
  kable()
```

The final `kable()` call turns the result into a formatted table. **The `tbl-` prefix makes the labelled table cross-referenceable.** Edit the caption if needed, then render again.

3.8 Add a discussion and cross-references

Under **Discussion**, write a short paragraph describing the pattern in the figure and the differences among the regions. Refer to both results using their labels. For example:

```
As shown in @fig-cases, cases peaked mid-year (see @tbl-summary for totals).
```

Render the document. Quarto should replace both labels with numbered, clickable references.

4 Bonus

4.1 Add a bibliography

1. Add `bibliography: references.bib` to the YAML header.
2. Cite the source with `[@li2025malaria]` in a sentence in your discussion.
3. Render the report. Quarto will add a References section automatically.

Open `references.bib` if you would like to inspect the bibliography entry and its citation key.

4.2 Apply the Swiss TPH report format

The workshop folder already contains the Swiss TPH Quarto extension. Replace your current `format` setting with:

```
format:
  report-typst:
    section-numbering: "1.1.1"
```

Render again. The report content is unchanged, but the extension applies the Swiss TPH layout, logo, colours, and typography.

5 Completion checklist

Your finished report should contain:

- a title and your name;
- an introduction with Markdown formatting and inline R results;
- a captioned line chart;
- a captioned summary table;
- a discussion with working figure and table cross-references; and
- optionally, a citation or the Swiss TPH report format.